Project 2: Turbojet Engine Design/Off-Design Performance Analysis

Riley J. Kyle*
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

Dr. David W. Riggins†
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude. At the design point the thrust is 306 N and the fuel flow rate is 0.0087 kg/s. Using temperature and pressure data at each trajectory point as well as a linear variation in Mach number and fuel throttle setting the design space was found numerically using a code developed in python. These results were then bound by the maximum and minimum values allowed by the engine control system. Compressor ratios higher than 18.0 or lower than 7.5, shaft RPM greater than 66000, and turbine entrance total temperatures higher than 1800K were removed, as well as some non-physical results.

Nomenclature

T = temperature $[K]$
P = pressure $[Pa]$
ρ = density $\left[kg/m^3\right]$
T_t = total temperature $[K]$
P_t = total pressure $[Pa]$
τ = $\frac{T_{t,out}}{T_{t,in}}$
π = $\frac{P_{t,out}}{P_{t,in}}$
η = isentropic efficiency
u = velocity $[m/s]$
M = mach number
A = area $\left[m^2\right]$
F = thrust $[N]$

Subscripts

d = diffuser

*Aerospace Engineering Student, MAE
†Professor of Mechanical and Aerospace Engineering, MAE

c = compressor

b = burner

t = turbine

n = nozzle

I. Introduction

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude. Temperature and pressure date was only given for the 8000 m case so standard atmosphere curve fits were used for the other trajectory points. The temperature and pressure information is shown in table 1. This data was used when running the solver at each altitude.

Table 1 Freestream Conditions

Altitude	T_0 (K)	P_0 (Pa)
Sea Level	288.15	101325
4000 m	262.15	61640
8000 m	236.23	35651
12000 m	216.65	19330

In this project we will be utilizing the assumption of the working fluid being a calorically perfect gas, air, with a constant C_p, γ, and R of 1004.5 J/kgK, 1.4, and 287 J/kgK respectively. It is also stated that this is a FAT jet that is choked at the turbine face and nozzle throat. This tells us that A_4, A_8, τ_t, and π_t are constant across the design space. Additionally, it was assumed that the fuel flow rate is small, and the turbine efficiency does not change; As shown in previous assignments, even without this assumption the results only change slightly.

The compressor isentropic efficiency and corrected rpm are given by the following equations:

$$\eta_c = -2.3219 + 1.7094\left(\frac{T_{t4}}{T_{t2}}\right) - 0.2897\left(\frac{T_{t4}}{T_{t2}}\right)^2 + 0.0155\left(\frac{T_{t4}}{T_{t2}}\right)^3 \tag{1}$$

$$\eta_c = 22352.47 + 3718.445\pi_c - 52.5576\pi_c^2 - 1.0899\pi_c^3 \tag{2}$$

II. Completely define the on-design engine fluid dynamics and performance

A. Methodology

The given engine information for the on design point is shown in table 2. Also, for this case we are to assume that there us full mass capture. Using these values and the equations presented in this section we can completely define the on-design condition.

Table 2 On Design

Altitude	8000	m
M_0	0.85	
T_0	236.23	K
P_0	35651	Pa
M_2	0.5	
A_1	0.00318	m^2
π_c	14.43	
π_b	1.0	
π_c	0.989163	
η_c	0.90	
T_{t4}	1500	K

The equations used for the engine analysis are:

$$P = \rho RT \tag{3}$$

$$u = M\sqrt{\gamma RT} \tag{4}$$

$$C_p T_t = C_p T + \frac{u^2}{2} \tag{5}$$

$$\dot{m} = \rho u A \tag{6}$$

$$\frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2 \tag{7}$$

$$\frac{P_t}{P} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{8}$$

$$F = \dot{m}\left(u_8 - u_0\right) + \left(P_8 - P_0\right) A_8 \tag{9}$$

The Python code for this section is shown in Appendix A

B. Results

The completely defined on-design engine is shown in table 3 and 4. Station 6, the isentropic duct, and station 7, the converging section of the nozzle are not tabulated here.

Table 3 On-Design Operation

		Station						
		0	1	2	3	4	5	8
u	m/s	261.873	261.873	160.826	160.826	708.696	566.957	621.38
P	Pa	35651	35651	47679.6	758088	431148	171131	152490
T	K	236.23	236.23	257.491	604.339	1250	993.151	960.96
M		0.85	0.85	0.5	0.32637	1	0.897506	1
ρ	kg/m^3	0.525841	0.525841	0.645192	4.37076	1.20181	0.600386	0.55291
A	m^2	0.00318	0.00318	0.00422016	0.000622961	0.000514136	0.00128645	0.00127456
P_t	Pa	57177.7	57177.7	56558.1	816133	816133	288653	288653
T_t	K	270.365	270.365	270.365	617.214	1500	1153.15	1153.15

Table 4 Quantities of Interest

τ_t		0.7687676
π_t		0.3536833
F	N	306.34619
$\dot{m}_{air}$	kg/s	0.4378983
$\dot{m}_{fuel}$	kg/s	0.0087261
f		0.0199272

III. Develop the complete performance envelope for this specific turbojet engine

A. Methodology

In a separate Python file, shown in Appendix B, the date that would remain constant were carried over. The constant values were pi_t, τ_t, η_t, π_d, π_b, $\dot{m}_f$, A_1, A_4, A_8, M_4, and M_8. Additionally, the fuel flow rate for the on-design condition was defined as the 100% fuel throttle setting.

First, a mesh of the test points was created for each altitude. This contained two square matrices, one with Mach number, and the other with the fuel throttle setting; This ensures that I match every Mach number with every fuel throttle setting and also improves run times by reducing the number of loops required. Figure 1 shows what this looks like. These matrices were flattened into a 1D array to simplify the later loop

Following the generation of the sample points a for loop was initialized to calculate the performance at each point. Using a sample Mach number and known inlet conditions values were calculated up to the compressor face in the order shown below. Here u_0, τ_r, π_r, T_{t0}, P_{t0}, T_{t2}, and P_{t2} were calculated. The mass flow if there was no spillage was also

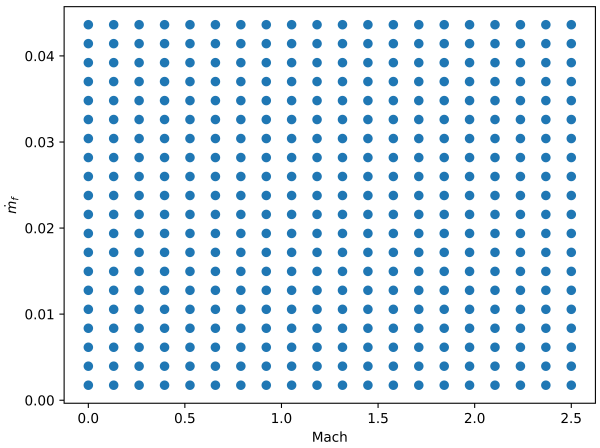


Fig. 1 Evaluation points example: 20x20

calculated. This is used in later calculations to find the loss in mass flow through the engine. This is shown in equation 10. The equation for spillage is shown in equation 11.

$$\dot{m}_{NS} = \rho_0 u_0 A_1 \tag{10}$$

$$Spillage = \dot{m}_{NS} - \dot{m}_{actual} \tag{11}$$

Since we are lacking knowledge of the performance of the compressor and turbo-machinery at this sample point we need to derive a procedure for numerically solving for this. In doing this I found that a slight modification to the procedure from homework 3 worked. The following equations were used to create a function of only the total temperature at the turbine face.

Compressor temperature ratio:

$$\tau_c = 1 + \frac{T_{t4}}{T_{t0}} (1 - \tau_t) \tag{12}$$

Compressor pressure ratio:

$$\pi_c = (1 + \eta_c (\tau_c - 1))^{\frac{\gamma}{\gamma - 1}} \tag{13}$$

η_c is also a function of T_{t4} as shown in equation 1.

Actual mass flow of air:

$$\dot{m} = \frac{\Gamma}{\sqrt{R T_{t4}}} A_4 \left[P_0 \pi_r \pi_d \pi_c \pi_b \right] \tag{14}$$

These were all combined into the following equation. Since we picked the fuel flow rate of this current iteration we

can write the following equation.

$$0 = \frac{\dot{m} C_p}{h} \left(T_{t4} - \tau_c T_{t2} \right) - \dot{m}_f \tag{15}$$

Since this is multiple equations all with the same unknown variable we can use a root solver to find T_{t4}. In this case I am using the python function scipy.optimize.fsolve(). Here I would enter the equation, equation 15 in this case, and initial guess for the temperature. For each case except for the 12 km case I used the initial guess of 900 K, however I noticed the 12 km case was very sensitive to this initial guess. For this I ended up using 700 K. After solving for this value the rest of the engine can be solved for. Using the π_c calculated from equation 13 we can use equation 2 to find the turbo-machinery RPM. and since the turbine temperature and pressure ratios are constant we can use those to directly calculate the total pressure and temperature at the exit.

$$T_{t8} = T_{t4} \tau_t \tag{16}$$

$$P_{t8} = P_{t4} \pi_t \tag{17}$$

Then using isentropic mach relations we can find the static pressure and temperature at the exit. For this analysis we are assuming that the nozzle exit/throat is choked:

$$T_8 = \frac{T_{t8}}{1 + \frac{\gamma - 1}{2}} \tag{18}$$

$$P_8 = \frac{P_{t8}}{\left(1 + \frac{\gamma - 1}{2} \right)^{\frac{\gamma}{\gamma - 1}}} \tag{19}$$

This can then be used to calculate the exit velocity:

$$u_8 = \sqrt{\gamma R T_8} \tag{20}$$

Now that we know the inlet exit velocity and pressure as well as the mass flow of air we can calculate the thrust of the engine at the current iteration:

$$F = \dot{m} \left(u_8 - u_0 \right) + \left(P_8 - P_0 \right) A_8 \tag{21}$$

B. Results

Figure 2-5 show the contours for the operational space for the engine at each sea level, 4000 m, 8000 m, and 12000 m. These results were then bound by the maximum and minimum values allowed by the engine control system. Compressor ratios higher than 18.0 or lower than 7.5, shaft RPM greater than 66000, and turbine entrance total temperatures higher than 1800K were removed, as well as some non-physical results. The numerical results were verified by using the on design point. At 8000 m this method gave the same solution as the method from part 1 for the design point. The turbo-machinery RPM contour color bar shows that it goes up to 67500 RPM because of the step size between contours and not because there are values above 66000 RPM.

IV. Conclusion

Using the process in this paper the design space of an engine can be analyzed if given curve fits for compressor efficiency and turbo-machinery RPM. For the given engine it can be seen that as the altitude increases, the engine needs less fuel to maintain a higher Mach number; The engine is also able to achieve higher mach numbers that what were possible at lower altitudes. An issue that presents itself with the increase in altitude is that the performance space allowed by the engine control system continues to decrease in size. This means that at higher altitudes the engine needs to fit within tighter and tighter margins to have an acceptable turbo-machinery RPM, compressor pressure ratio, and turbine temperature. There will likely be an altitude where it is not possible for the turbine to spool enough to maintain the necessary pressure for combustion.

Increasing altitude dec

There were many valid assumptions used in this analysis though there are certainly things that could be improved. For this analysis a massive contributor to error could be the assumption that the pressure ratio of the inlet is constant. As the engine goes through transonic and supersonic speeds one should expect to see shocks forming at the inlet which will cause significant changes in that pressure term. This pressure drop could probably be modeled using an 2D fit of Mach number and angle of attack. While the change will most likely be very small, accounting for the change in turbine efficiency would further improve accuracy. Accounting for variations in burner efficiency and pressure ratio could also have an impact on the results. Between Mach 0 and 2.5 there is a large difference in the inlet total pressure which would have an effect on the combustion efficiency. Having more information about the engine can allow for more robust mission planing for this engine which can then be integrated into the engine control system to allow for improved performance at different altitude and Mach numbers.

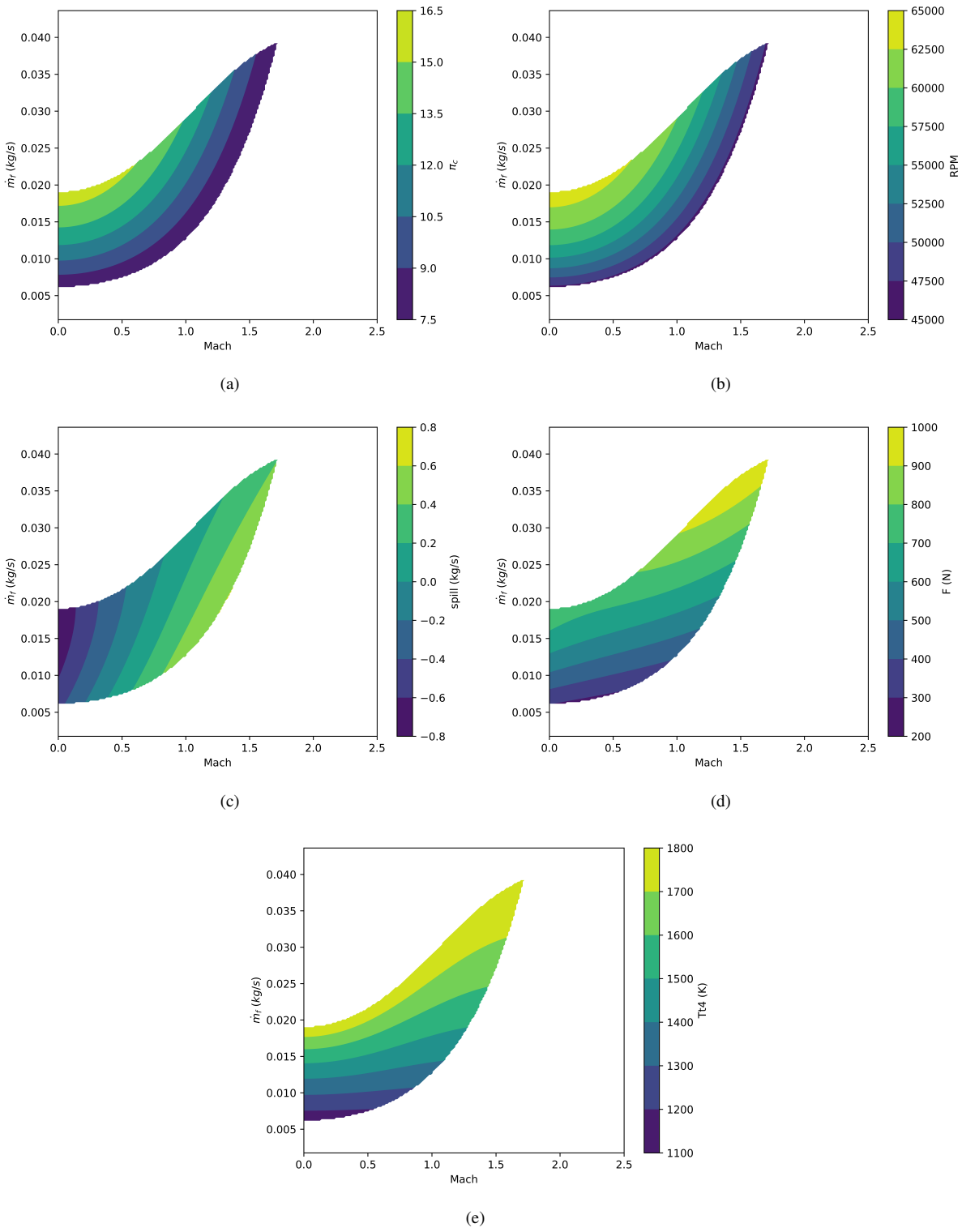


Fig. 2 Sea Level (a) π_c (b) Turbo-Machine RPM (c) Spillage (d) Thrust (e) T_{t4}

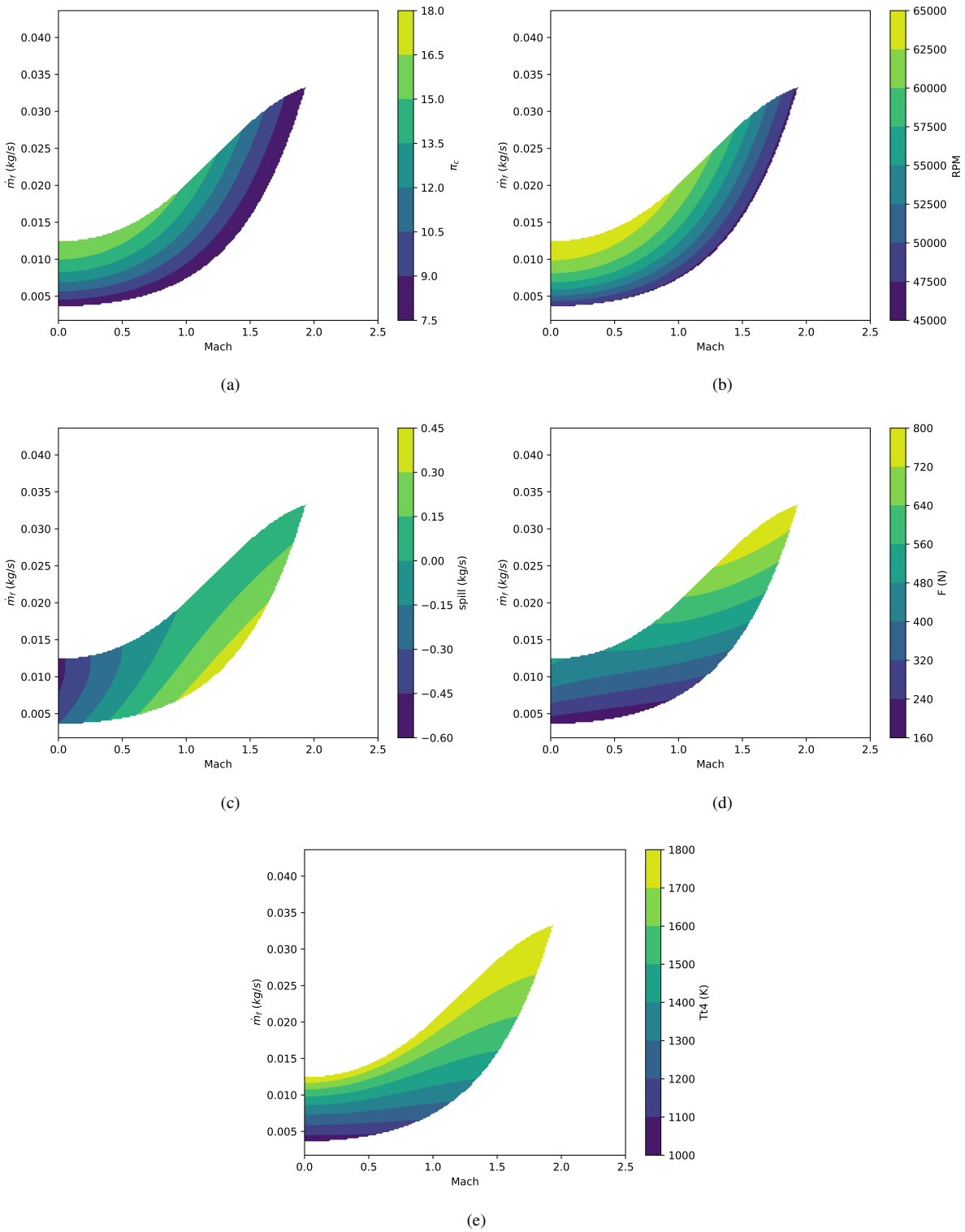


Fig. 3 4000 m (a) π_c (b) Turbo-Machine RPM (c) Spillage (d) Thrust (e) T_{t4}

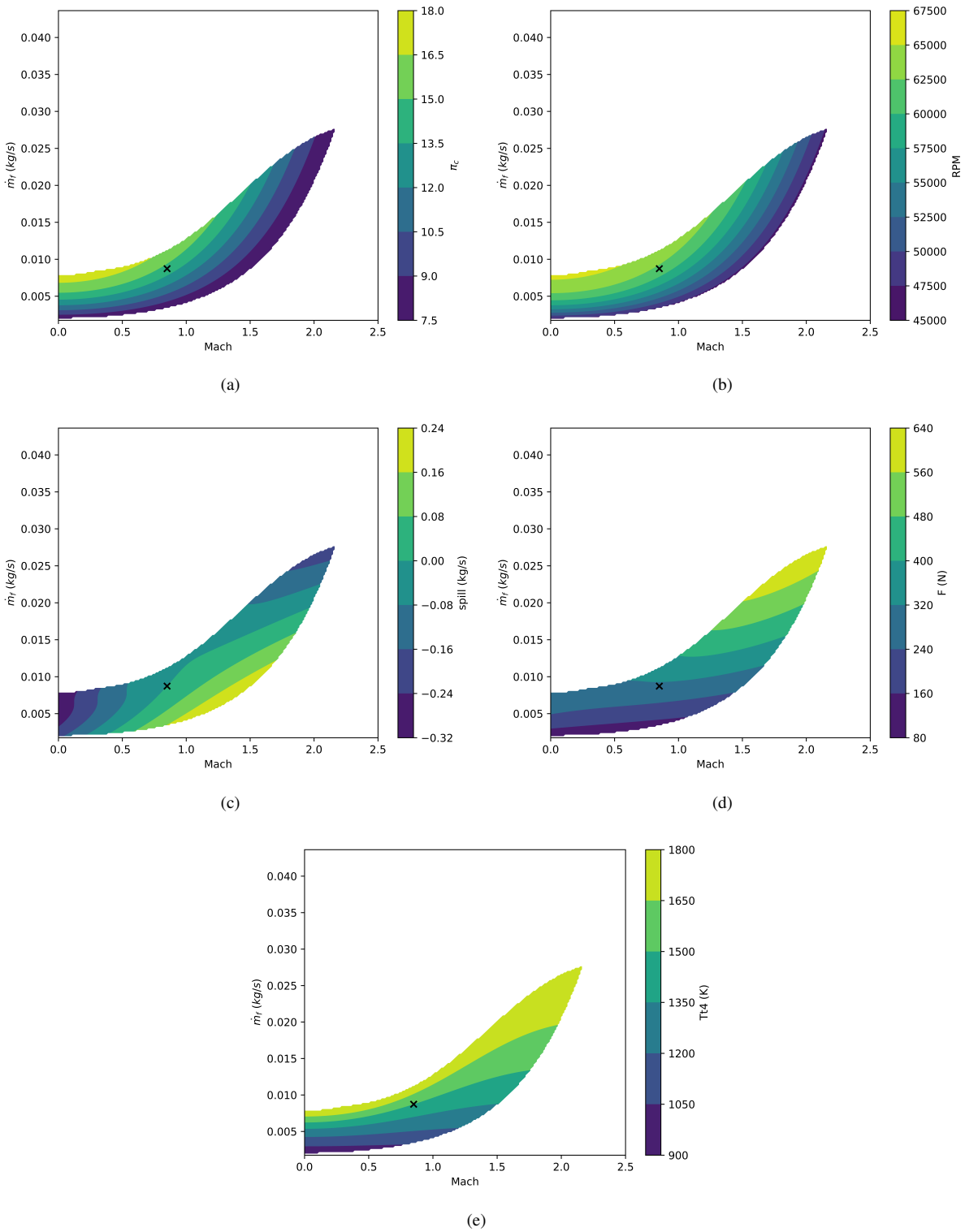


Fig. 4 8000 m (a) π_c (b) Turbo-Machine RPM (c) Spillage (d) Thrust (e) T_{t4}

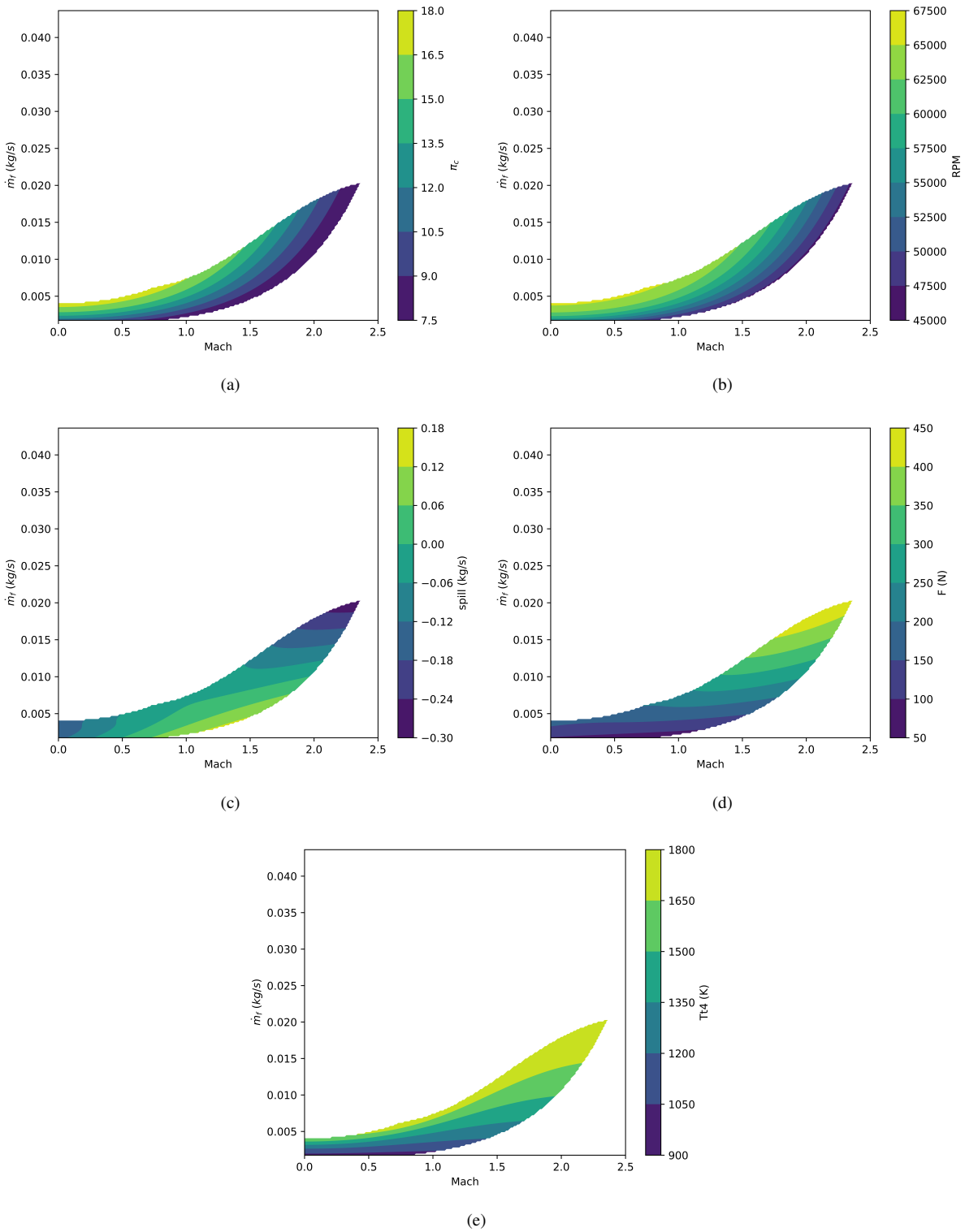


Fig. 5 12000 m (a) π_c (b) Turbo-Machine RPM (c) Spillage (d) Thrust (e) T_{t4}

V. Appendix A

```
    # A comment
    # %%
import numpy as np
def eta_c_fit(Tt2,Tt4):
    x = Tt4/Tt2
    a = -2.3219
    b = 1.7094
    c = -0.2897
    d = 0.0155
    return a + b*x + c*x**2 + d*x**3

def corr_rpm_fit(pi_c):
    x = pi_c
    aa = 22352.47
    bb = 3718.445
    cc = -52.5576
    dd = -1.0899
    return aa + bb*x + cc*x**2 + dd*x**3

gamma = 1.4
Cp = 1004.5
R = 287
h = 4.45e+7

class given:
    M0 = 0.85
    T0 = 236.23 # K
    P0 = 35651 # Pa
    M2 = 0.5
    A1 = 0.00318 # m^2
    A0 = A1

    pi_c = 14.43
    pi_b = 1.0
    pi_d = 0.989163
    eta_t = 0.90

    Tt4 = 1500 # K

u = np.zeros(9)
```

```
P = np.zeros(9)
T = np.zeros(9)
M = np.zeros(9)
rho = np.zeros(9)
A = np.zeros(9)
Pt = np.zeros(9)
Tt = np.zeros(9)

M[0] = given.M0
T[0] = given.T0
P[0] = given.P0
rho[0] = P[0]/(R*T[0])
A[0] = given.A0

u[0] = M[0]*np.sqrt(gamma*R*T[0])

tau_r = 1+(gamma-1)/2 * M[0]**2
pi_r = tau_r**(gamma/(gamma-1))

Tt[0] = T[0] * tau_r
Pt[0] = P[0] * pi_r

m_dot = rho[0] * u[0] * A[0]
m_dot

pi_r

M[1] = given.M0
T[1] = given.T0
P[1] = given.P0
rho[1] = P[1]/(R*T[1])
A[1] = given.A1

u[1] = M[1]*np.sqrt(gamma*R*T[1])



Tt[1] = Tt[0]
Pt[1] = Pt[0]

M[2] = given.M2
```

```
tau_d = 1
pi_d = given.pi_d

Tt[2] = Tt[1] * tau_d
Pt[2] = Pt[1] * pi_d

T[2] = Tt[2]/(1+(gamma-1)/2 * M[2]**2)
P[2] = Pt[2]/(1+(gamma-1)/2 * M[2]**2)**(gamma/(gamma-1))

rho[2] = P[2]/(R*T[2])

u[2] = M[2]*np.sqrt(gamma*R*T[2])

A[2] = m_dot / (rho[2] * u[2])

u[3] = u[2]

pi_c = given.pi_c
eta_c = eta_c_fit(Tt[2],given.Tt4)
tau_c =  ((pi_c**((gamma-1)/gamma)-1)/eta_c)+1

Tt[3] = Tt[2] * tau_c
Pt[3] = Pt[2] * pi_c

T[3] = (Cp*Tt[3] - u[3]**2/2)/Cp
M[3] = u[3]/np.sqrt(gamma*R*T[3])
P[3] = Pt[3]/(1+(gamma-1)/2 * M[3]**2)**(gamma/(gamma-1))

rho[3] = P[3]/(R*T[3])

A[3] = m_dot / (rho[3] * u[3])

tau_c

M[4] = 1
Tt[4] = given.Tt4

pi_b = given.pi_b

Pt[4] = Pt[3] * pi_b
```

```
T[4] = Tt[4]/(1+(gamma-1)/2 * M[4]**2)
P[4] = Pt[4]/(1+(gamma-1)/2 * M[4]**2)**(gamma/(gamma-1))

rho[4] = P[4]/(R*T[4])

u[4] = M[4]*np.sqrt(gamma*R*T[4])

A[4] = m_dot / (rho[4] * u[4])

tau_lambda = Tt[4] / T[0]

m_dot_f = m_dot * Cp * (Tt[4]-Tt[3]) / h

u[5] = 0.8 * u[4]

eta_t = given.eta_t

tau_t = 1-(tau_c - 1)*(tau_r/tau_lambda)
pi_t = (1-1/eta_t*(1-tau_t))**(gamma/(gamma-1))

Tt[5] = Tt[4] * tau_t
Pt[5] = Pt[4] * pi_t

T[5] = (Cp*Tt[5] - u[5]**2/2)/Cp
M[5] = u[5]/np.sqrt(gamma*R*T[5])
P[5] = Pt[5]/(1+(gamma-1)/2 * M[5]**2)**(gamma/(gamma-1))

rho[5] = P[5]/(R*T[5])

A[5] = m_dot / (rho[5] * u[5])

M[8] = 1

tau_n = 1
pi_n = 1

Tt[8] = Tt[5] * tau_n
Pt[8] = Pt[5] * pi_n

T[8] = Tt[8]/(1+(gamma-1)/2 * M[8]**2)
P[8] = Pt[8]/(1+(gamma-1)/2 * M[8]**2)**(gamma/(gamma-1))
```

```
rho[8] = P[8]/(R*T[8])

u[8] = M[8]*np.sqrt(gamma*R*T[8])

A[8] = m_dot / (rho[8] * u[8])

F = (m_dot) * (u[8] - u[0]) + (P[8] - P[0]) * A[8]
f = m_dot_f/m_dot

table = np.array([
    u,
    P,
    T,
    M,
    rho,
    A,
    Pt,
    Tt
])
```

VI. Appendix B

```
import numpy as np
import scipy
import matplotlib.pyplot as plt

# %%
pi_t = 0.35368326100748115
tau_t = 0.7687676014665914
eta_t = 0.90
pi_d =  0.989163
pi_b = 1

m_dot_fR = 0.008726071272118822

A1 = 0.00318
A4 = 0.0005141364620689669
A8 = 0.0012745633749949147
M4 = 1
M8 = 1

# %%
tau_d = 1

# %%
pi_c_max = 18
pi_c_min = 7.5

rpm_max = 66000

Tt4_max = 1800

# %%
gamma = 1.4
Gamma = (2/(gamma+1))**((gamma+1)/(2*(gamma-1)))*np.sqrt(gamma)
Cp = 1004.5
R = 287
h = 4.45e+7

Alt_list = np.array([0,4000,8000,12000])
T0_list = np.array([288.15 ,262.15 ,236.23 ,216.65])
P0_list = np.array([101325 , 61640.235,35651 ,19330.405])
```

```
rho0_list = P0_list/(R*T0_list)
a0_list = np.sqrt(gamma*R*T0_list)

# %%
n = 200
Alt_size = Alt_list.size

M0 = np.linspace(0,2.5,n)
m_dot_f = np.linspace(.2,5,n) * m_dot_fR

M0, m_dot_f = np.meshgrid(M0,m_dot_f)

M0 = np.matrix.flatten(M0)
m_dot_f = np.matrix.flatten(m_dot_f)

#M0 = np.array([0.85])
#m_dot_f =  np.array([m_dot_fR])

j = 3

Alt = Alt_list[j]
T0 = T0_list[j]
P0 = P0_list[j]
rho0 = P0/(R*T0)
a0 = np.sqrt(gamma*R*T0)



Tt4_plot = np.zeros(n**2)
rpm_plot = np.zeros(n**2)
pi_c_plot = np.zeros(n**2)
Thrust_plot = np.zeros(n**2)
spill_plot = np.zeros(n**2)

def eta_c_fit(Tt4):
        x = Tt4/Tt2
        a = -2.3219
        b = 1.7094
        c = -0.2897
        d = 0.0155
        return a + b*x + c*x**2 + d*x**3
```

```
def corr_rpm_fit(pi_c):
    x = pi_c
    aa = 22352.47
    bb = 3718.445
    cc = -52.5576
    dd = -1.0899
    return aa + bb*x + cc*x**2 + dd*x**3

def _tau_c(Tt4):
    return 1+(Tt4/Tt0)*(1-tau_t)

def _pi_c(Tt4):
    return  (1+eta_c_fit(Tt4)*(_tau_c(Tt4)-1))**(gamma/(gamma-1))

def _m_dot(Tt4):
    return Gamma / np.sqrt(R) * A4 * (P0*pi_r*pi_d*_pi_c(Tt4)*pi_b)*1/(np.sqrt(Tt4))

def _mdotf(Tt4):
    return _m_dot(Tt4)*Cp*(Tt4 - _tau_c(Tt4) * Tt2)/h-m_dot_f[i]

for i in range(M0.size):
    u0 = M0[i] * np.sqrt(gamma*R*T0)

    m_dot_no_spill = rho0 * u0 * A1

    tau_r = 1+(gamma-1)/2 * M0[i]**2
    pi_r = tau_r**(gamma/(gamma-1))

    Tt0 = T0 * tau_r
    Pt0 = P0 * pi_r

    Tt2 = Tt0
    Pt2 = Pt0 * pi_d

    Tt4_guess = 800
    Tt4 = scipy.optimize.fsolve(_mdotf,Tt4_guess,maxfev=90000)[0]

    tau_c = _tau_c(Tt4)
    pi_c = _pi_c(Tt4)

    m_dot_air = _m_dot(Tt4)
```

```
    corr_rpm = corr_rpm_fit(pi_c)# np.sqrt(Tt0/T0)

    eta_c = eta_c_fit(Tt4)



    Tt8 = Tt4 * tau_t
    Pt8 = Pt2 * pi_c * pi_b * pi_t

    T8 = Tt8 / (1 + (gamma-1)/2 * M8**2)
    P8 = Pt8 / (1 + (gamma-1)/2 * M8**2)**(gamma/(gamma-1))

    u8 = M8 * np.sqrt(gamma*R*T8)

    Tt4_plot[i] = Tt4
    rpm_plot[i] = corr_rpm
    pi_c_plot[i] = pi_c
    spill_plot[i] = (m_dot_air-m_dot_no_spill)
    Thrust_plot[i] = (m_dot_air) * (u8 - u0) + (P8 - P0) * A8

    if Tt4 > Tt4_max or Tt4 <= Tt0 or Tt4 == Tt4_guess or pi_c > pi_c_max or pi_c < pi_c_min or
                                                    corr_rpm > rpm_max:
        Tt4_plot[i] = np.nan
        pi_c_plot[i] = np.nan
        rpm_plot[i] = np.nan
        Thrust_plot[i] = np.nan
        spill_plot[i] = np.nan
        continue

Tt4_plot = np.reshape(Tt4_plot,(n,n))
rpm_plot = np.reshape(rpm_plot,(n,n))
pi_c_plot = np.reshape(pi_c_plot,(n,n))
spill_plot = np.reshape(spill_plot,(n,n))
Thrust_plot = np.reshape(Thrust_plot,(n,n))
M0 = np.reshape(M0,(n,n))
m_dot_f = np.reshape(m_dot_f,(n,n))

fig = plt.figure(1)
ax = plt.axes()
cp = ax.contourf(M0,m_dot_f,Tt4_plot)
cbar = fig.colorbar(cp)
cbar.ax.set_ylabel(r'Tt4 (K)')
```

```
if j == 2:
    ax.scatter(0.85,m_dot_fR,c='k',marker='x')
ax.set_xlabel('Mach')
ax.set_ylabel(r'$\dot{m}_f \ (kg/s)$')
plt.tight_layout()
plt.savefig(r'{}_m_Tt4.png'.format(Alt_list[j]), dpi=1200)
plt.show()

# %%
fig = plt.figure(2)
ax = plt.axes()
cp = ax.contourf(M0,m_dot_f,rpm_plot,cmap='viridis')

if j == 2:
    ax.scatter(0.85,m_dot_fR,c='k',marker='x')
ax.set_xlabel('Mach')
ax.set_ylabel(r'$\dot{m}_f \ (kg/s)$')
plt.tight_layout()
cbar = fig.colorbar(cp)
cbar.ax.set_ylabel('RPM')
plt.savefig(r'{}_m_rpm.png'.format(Alt_list[j]), dpi=1200)
plt.show()

# %%
fig = plt.figure(3)
ax = plt.axes()
cp = ax.contourf(M0,m_dot_f,pi_c_plot)

if j == 2:
    ax.scatter(0.85,m_dot_fR,c='k',marker='x')
ax.set_xlabel('Mach')
ax.set_ylabel(r'$\dot{m}_f \ (kg/s)$')
plt.tight_layout()
cbar = fig.colorbar(cp)
cbar.ax.set_ylabel(r'$\pi_c$')
plt.savefig(r'{}_m_pic.png'.format(Alt_list[j]), dpi=1200)
plt.show()

# %%
fig = plt.figure(4)
```

```
ax = plt.axes()
cp = ax.contourf(M0,m_dot_f,spill_plot)

if j == 2:
    ax.scatter(0.85,m_dot_fR,c='k',marker='x')
ax.set_xlabel('Mach')
ax.set_ylabel(r'$\dot{m}_f \ (kg/s)$')
plt.tight_layout()
cbar = fig.colorbar(cp)
cbar.ax.set_ylabel(r'spill (kg/s)')
plt.savefig(r'{}_m_spill.png'.format(Alt_list[j]), dpi=1200)
plt.show()

# %%
fig = plt.figure(5)
ax = plt.axes()
cp = ax.contourf(M0,m_dot_f,Thrust_plot)

if j == 2:
    ax.scatter(0.85,m_dot_fR,c='k',marker='x')
ax.set_xlabel('Mach')
ax.set_ylabel(r'$\dot{m}_f \ (kg/s)$')
plt.tight_layout()
cbar = fig.colorbar(cp)
cbar.ax.set_ylabel(r'F (N)')
plt.savefig(r'{}_m_thrust.png'.format(Alt_list[j]), dpi=1200)
plt.show()
```